BABU BANARSI DAS UNIVERSITY LUCKNOW

BBD UNIVERSITY

SCHOOL OF COMPUTER APPLICATION

GROUP PROJECT ON BIG DATA FUNDAMENTALS

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PROJECT :- ANIME RECOMMENDATION ANALYSIS

BCADS25

GROUP-B

ANIME RECOMMENDATION ANALYSIS

- **Architecture: This project follows a Big Data pipeline using Hadoop ecosystem.**

- **Data Flow:**

Raw Dataset → HDFS → MapReduce Processing → Processed Data → Hive → Insights

- **Explanation:**

- **Dataset is stored in HDFS**
- **MapReduce processes the data**
- **Processed data is analyzed using Hive**
- **Insights are generated**

➢ Tools Used:

- **HDFS**
- **MapReduce**
- **Hive**
- **Python**

➢ Technologies:

- **CSV Dataset**
- **Hadoop Ecosystem**
- **Python Programming Use of Generative AI in the Project AI tools were used to assist in development and understanding.**

➢ Areas Where AI Was Used:

1. **Code Generation**
2. **Debugging**
3. **Query Writing**
4. **Concept Understanding**

➢ **Benefits:**

- **Faster development**
- **Reduced errors**
- **Better understanding**
- **Improved efficiency**

➢ **Validation: All AI-generated content was reviewed and verified. Dataset Description.**

The dataset used is a food orders dataset in CSV format.

➢ **Columns:**

- **UserID**
- **AnimeID**
- **Anime_Title**
- **Rating**
- **Year**
- **Episodes**
- **Studio**
- **Status**
- **ReviewScore**

- **Purpose: To efficiently analyze large anime rating data using Hadoop MapReduce and calculate average ratings for better insights and recommendations.**

- **Dataset Source: Online dataset for project use.**

- **Conclusion :- This project demonstrates the use of Hadoop MapReduce to efficiently process anime rating data and compute average ratings, providing useful insights and improving understanding of Big Data technologies.**

Step 1: Create Working Directory

```
[cloudera@quickstart ~]$ mkdir -p ~/anime_project/mapreduce
[cloudera@quickstart ~]$
```

Step 2: Go into that folder :

```
[cloudera@quickstart ~]$ cd ~/anime_project/mapreduce
[cloudera@quickstart mapreduce]$
```

Step 3: Create anime.csv file

```
[cloudera@quickstart mapreduce]$ nano anime.csv
[cloudera@quickstart mapreduce]$
```

Step 4: Paste the data

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
 GNU nano 2.0.9                                File: anime.csv

20,108,Fullmetal Alchemist,Action,5,2005,51,Bones,Completed,9
21,109,Code Geass,Mecha,5,2006,25,Sunrise,Completed,8
22,110,Bleach,Action,4,2004,366,Studio Pierrot,Completed,7
23,111,One Piece,Adventure,5,2000,1000,Toei Animation,Ongoing,7
24,112,Neon Genesis Evangelion,Mecha,4,1995,26,Gainax,Completed,8
25,113,Cowboy Bebop,Sci-Fi,5,1998,26,Sunrise,Completed,9
26,114,Trigun,Western,4,1998,26,Madhouse,Completed,8
27,115,Yu Yu Hakusho,Action,4,1992,112,Pierrot,Completed,8
28,113,Cowboy Bebop,Sci-Fi,5,1998,26,Sunrise,Completed,9
29,114,Trigun,Western,5,1998,26,Madhouse,Completed,8
30,115,Yu Yu Hakusho,Action,3,1992,112,Pierrot,Completed,8
```

Step 5: Create mapper.py file

```
[cloudera@quickstart ~]$ nano mapper.py
```

Step 6: Paste mapper code

```
Applications  Places  System
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
GNU nano 2.0.9                                File: mapper.py

#!/usr/bin/env python
import sys
for line in sys.stdin:
    if "userId" in line:
        continue
    try:
        data = line.strip().split(',')
        anime = data[2]
        rating = float(data[4])
        print("{}\t{}".format(anime, rating))
    except:
        pass
```

Step 7: Create reducer.py file

```
[cloudera@quickstart ~]$ nano reducer.py
```

Step 8: Paste reducer code

```
File  Edit  View  Search  Terminal  Help
GNU nano 2.0.9                                File: reducer.py

#!/usr/bin/env python
import sys
anime = None
total = 0
count = 0
for line in sys.stdin:
    try:
        a, r = line.strip().split('\t')
        r = float(r)
        if a == anime:
            total += r
            count += 1
        else:
            if anime:
                print("{}\t{:.2f}\t{}".format(anime, total/count, count))
            anime = a
            total = r
            count = 1
    except:
        pass
if anime:
    print("{}\t{:.2f}\t{}".format(anime, total/count, count))
```

Step 9: Make Scripts Executable

```
[cloudera@quickstart ~]$ chmod +x mapper.py reducer.py
```

Step 10: Test mapper locally

```
[cloudera@quickstart ~]$ cat anime.csv | python mapper.py
```

Step 11: Test full pipeline locally

```
[cloudera@quickstart ~]$ cat anime.csv | python mapper.py | sort | python reducer.py
```

Step 12: Verify you are in right folder

```
[cloudera@quickstart ~]$ pwd
/home/cloudera
```

Step 13: Create HDFS Directories

```
[cloudera@quickstart ~]$ hdfs dfs -mkdir -p /user/cloudera/animeproject/input
```

Step 14: Check HDFS Directories

```
[cloudera@quickstart ~]$ hdfs dfs -mkdir -p /user/cloudera/animeproject/output
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/animeproject/
Found 2 items
drwxr-xr-x   - cloudera cloudera          0 2026-04-25 05:02 /user/cloudera/animeproject/input
drwxr-xr-x   - cloudera cloudera          0 2026-04-25 05:04 /user/cloudera/animeproject/output
```

Step 15: Upload Dataset to HDFS

```
[cloudera@quickstart ~]$ hdfs dfs -put anime.csv /user/cloudera/animeproject/input/
put: `/user/cloudera/animeproject/input/anime.csv': File exists
```

Step 16: Run Mapreduce Job

```
[cloudera@quickstart ~]$ hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar -files mapper.py,reducer.py -input /user/cloudera/animeproject/input/anime.csv -output /user/cloudera/animeproject/output/job1 -mapper "python mapper.py"
-reducer "python reducer.py"
packageJobJar: [] [/usr/lib/hadoop-mapreduce/hadoop-streaming-2.6.0-cdh5.13.0.jar] /tmp/streamjob3345528255057190126.jar tmpDir=null
26/04/26 04:27:01 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/26 04:27:01 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/26 04:27:01 WARN security.UserGroupInformation: PriviledgedActionException as:cloudera (auth:SIMPLE) cause:org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory hdfs://quickstart.cloudera:8020/user/cloudera/animeproc
ject/output/job1 already exists
26/04/26 04:27:01 WARN security.UserGroupInformation: PriviledgedActionException as:cloudera (auth:SIMPLE) cause:org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory hdfs://quickstart.cloudera:8020/user/cloudera/animeproc
ject/output/job1 already exists
```

Step 17: Check Output

```
[cloudera@quickstart ~]$ hdfs dfs -cat /user/cloudera/animeproject/output/job1/part-00000
Attack on Titan 4.5     2
Bleach  3.0     1
Code Geass      5.0     1
Cowboy Bebop    5.0     1
Death Note      4.5     2
Demon Slayer    4.0     1
Fairy Tail      3.0     1
Fullmetal Alchemist     5.0     1
Naruto  3.5     2
Neon Genesis Evangelion 4.0     1
One Piece       4.5     2
Steins;Gate     5.0     1
Tokyo Ghoul     4.0     1
Trigun  4.0     1
Your Name       5.0     1
Yu Yu Hakusho   3.0     1
[cloudera@quickstart ~]$
```